

ASSIGNMENT NO.5: Aggregate Functions and Data Analysis

Student Name	Amit Panorekar	Roll Number	20968
Batch	SY. BSc Data Science	Topic	WB(Assignment-5)

Q1) Create a MySQL database named CollegeDB and a table named StudentResult.

The table should contain the following columns:

Column Name	Data Type	Constraint
StudentID	INT	PRIMARY KEY
StudentName	VARCHAR(50)	
Course	VARCHAR(30)	
City	VARCHAR(30)	
Marks	INT	
FeesPaid	DECIMAL(10,2)	

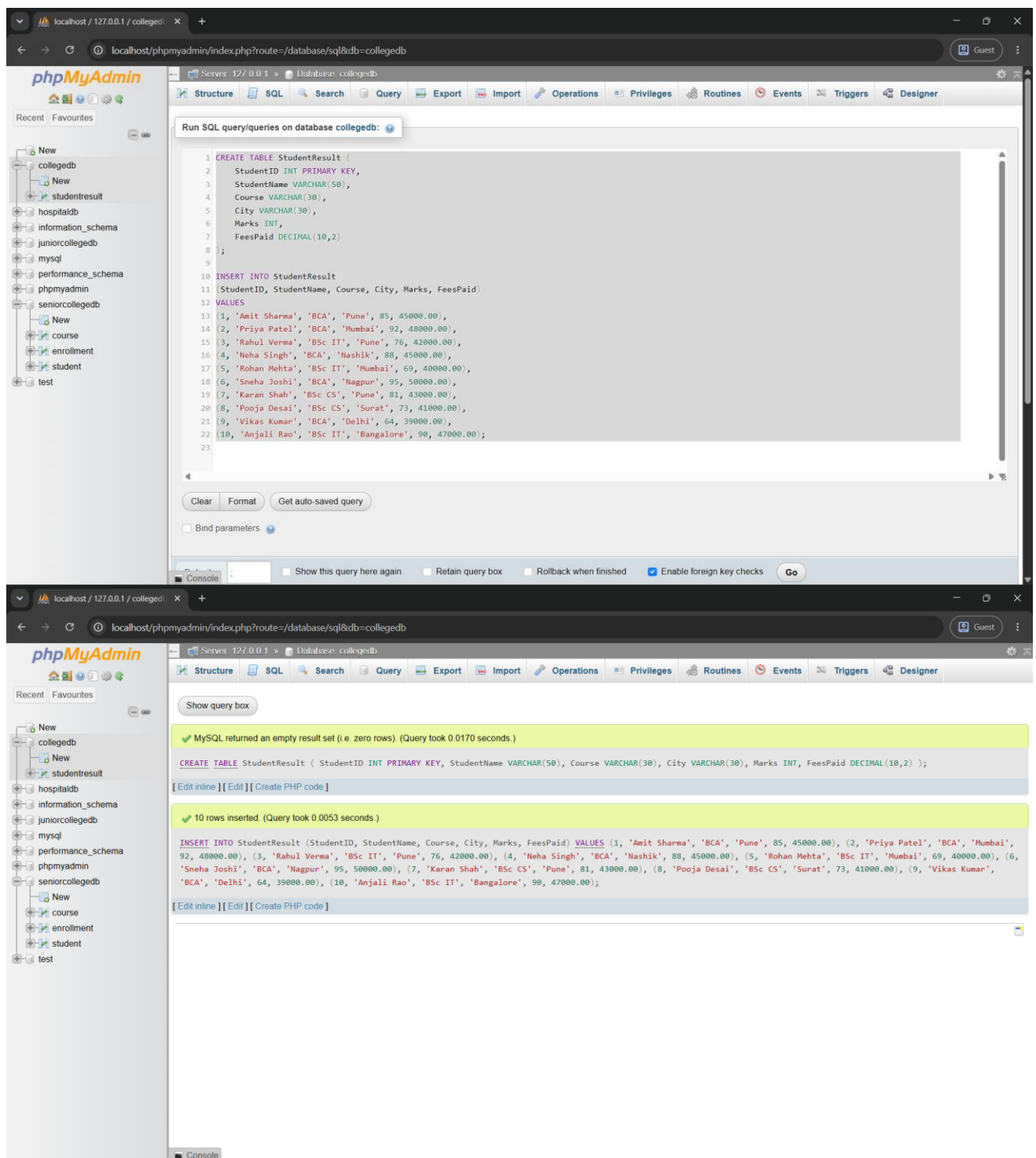
Insert at least 10 records into the table

QUERY:

```
CREATE TABLE StudentResult (  
    StudentID INT PRIMARY KEY,  
    StudentName VARCHAR(50),  
    Course VARCHAR(30),  
    City VARCHAR(30),  
    Marks INT,  
    FeesPaid DECIMAL(10,2)  
);
```

```
INSERT INTO StudentResult  
(StudentID, StudentName, Course, City, Marks, FeesPaid)  
VALUES  
(1, 'Amit Sharma', 'BCA', 'Pune', 85, 45000.00),  
(2, 'Priya Patel', 'BCA', 'Mumbai', 92, 48000.00),  
(3, 'Rahul Verma', 'BSc IT', 'Pune', 76, 42000.00),  
(4, 'Neha Singh', 'BCA', 'Nashik', 88, 45000.00),  
(5, 'Rohan Mehta', 'BSc IT', 'Mumbai', 69, 40000.00),  
(6, 'Sneha Joshi', 'BCA', 'Nagpur', 95, 50000.00),  
(7, 'Karan Shah', 'BSc CS', 'Pune', 81, 43000.00),  
(8, 'Pooja Desai', 'BSc CS', 'Surat', 73, 41000.00),  
(9, 'Vikas Kumar', 'BCA', 'Delhi', 64, 39000.00),  
(10, 'Anjali Rao', 'BSc IT', 'Bangalore', 90, 47000.00);
```

OUTPUT:



Q2) Write MySQL queries to find the following from the StudentResult table:

Requirement
Total number of students
Total fees paid
Average marks
Lowest marks
Highest marks

QUERY:

-- 1. Total number of students

```
SELECT COUNT(*) AS TotalStudents  
FROM StudentResult;
```

-- 2. Total fees paid

```
SELECT SUM(FeesPaid) AS TotalFeesPaid  
FROM StudentResult;
```

-- 3. Average marks

```
SELECT AVG(Marks) AS AverageMarks  
FROM StudentResult;
```

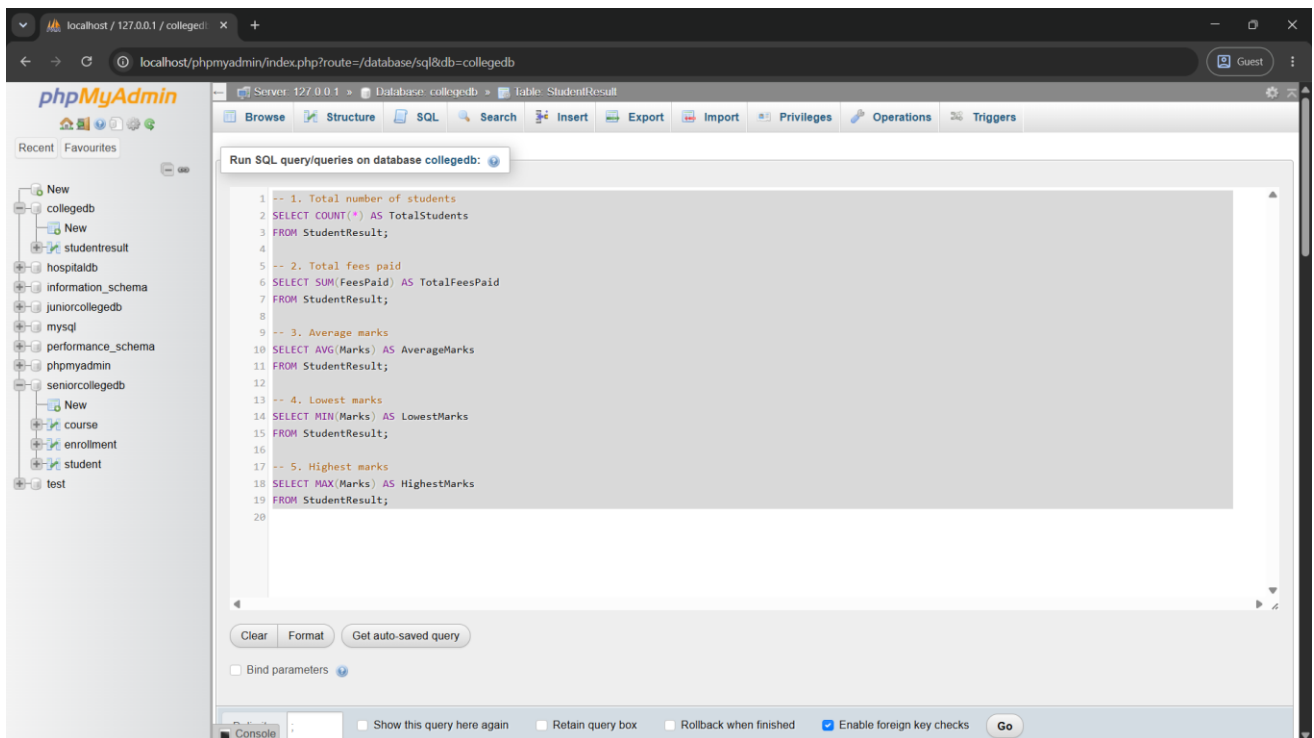
-- 4. Lowest marks

```
SELECT MIN(Marks) AS LowestMarks  
FROM StudentResult;
```

-- 5. Highest marks

```
SELECT MAX(Marks) AS HighestMarks  
FROM StudentResult;
```

OUTPUT:



localhost / 127.0.0.1 / collegedb

phpMyAdmin

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Server: 127.0.0.1 > Database: collegedb > Table: StudentResult

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Your SQL query has been executed successfully

-- 1. Total number of students `SELECT COUNT(*) AS TotalStudents FROM StudentResult;`

[Edit inline] [Edit] [Create PHP code]

Extra options

TotalStudents

10

Query results operations

Print Copy to clipboard Export Display chart Create view

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Showing rows 0 - 0 (1 total, Query took 0.0004 seconds)

-- 2. Total fees paid `SELECT SUM(FeesPaid) AS TotalFeesPaid FROM StudentResult;`

[Edit inline] [Edit] [Create PHP code]

Extra options

TotalFeesPaid

440000.00

Showing rows 0 - 0 (1 total, Query took 0.0003 seconds)

-- 3. Average marks `SELECT AVG(Marks) AS AverageMarks FROM StudentResult;`

[Edit inline] [Edit] [Create PHP code]

Extra options

AverageMarks

81.3000

Showing rows 0 - 0 (1 total, Query took 0.0003 seconds)

-- 4. Lowest marks `SELECT MIN(Marks) AS LowestMarks FROM StudentResult;`

[Edit inline] [Edit] [Create PHP code]

Extra options

LowestMarks

64

Console

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Server: 127.0.0.1 > Database: collegedb > Table: StudentResult

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Showing rows 0 - 0 (1 total, Query took 0.0003 seconds)

-- 3. Average marks `SELECT AVG(Marks) AS AverageMarks FROM StudentResult;`

[Edit inline] [Edit] [Create PHP code]

Extra options

AverageMarks

81.3000

Showing rows 0 - 0 (1 total, Query took 0.0003 seconds)

-- 4. Lowest marks `SELECT MIN(Marks) AS LowestMarks FROM StudentResult;`

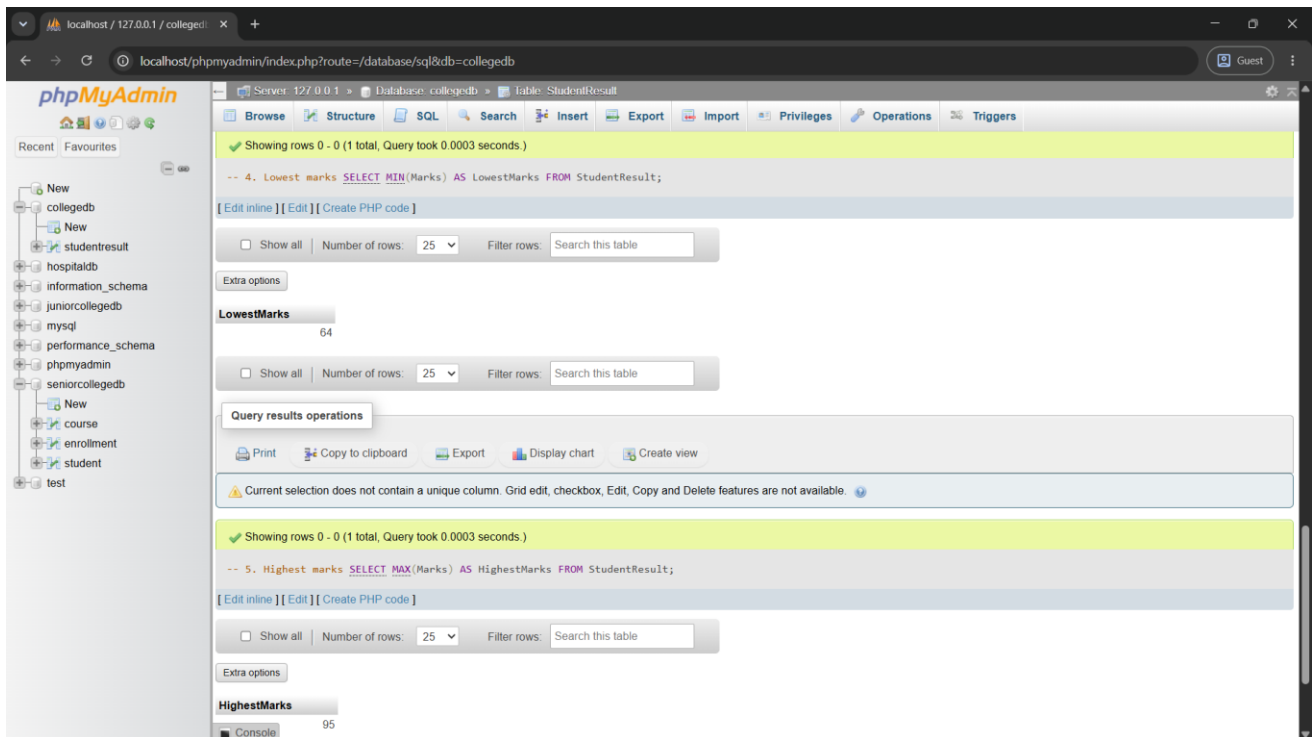
[Edit inline] [Edit] [Create PHP code]

Extra options

LowestMarks

64

Console



Q3) Write a MySQL query to count the number of students in each course.

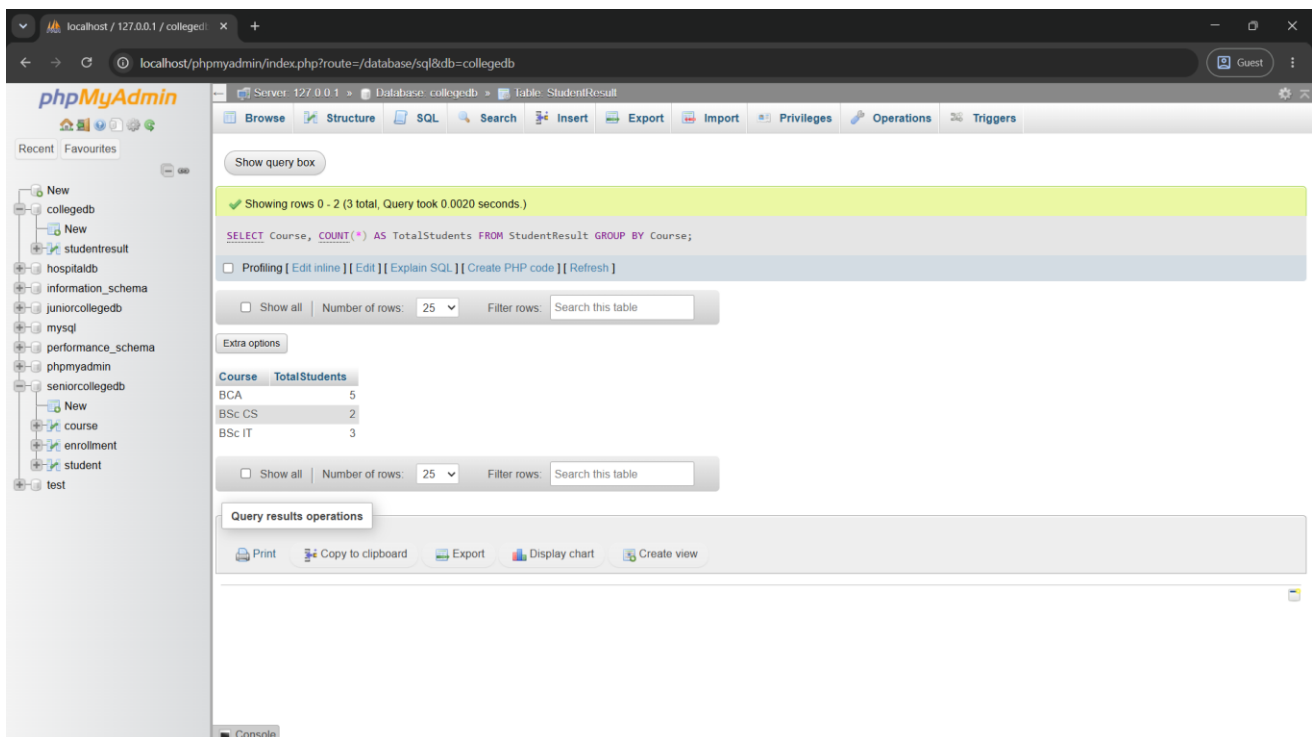
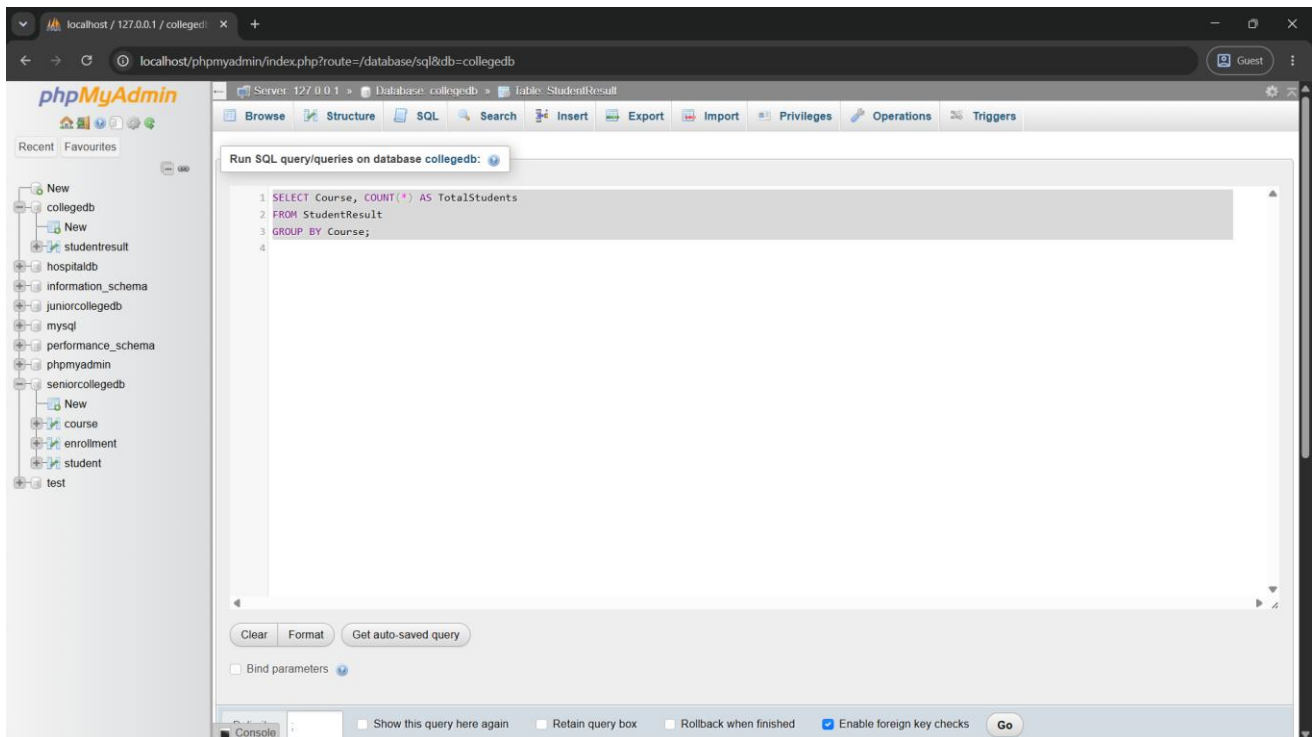
The output should contain:

Column
Course
TotalStudents

QUERY:

```
SELECT Course, COUNT(*) AS TotalStudents
FROM StudentResult
GROUP BY Course;
```

OUTPUT:



Q4) Write a MySQL query to find the average marks of students course-wise.

The output should contain:

Column
Course
AverageMarks

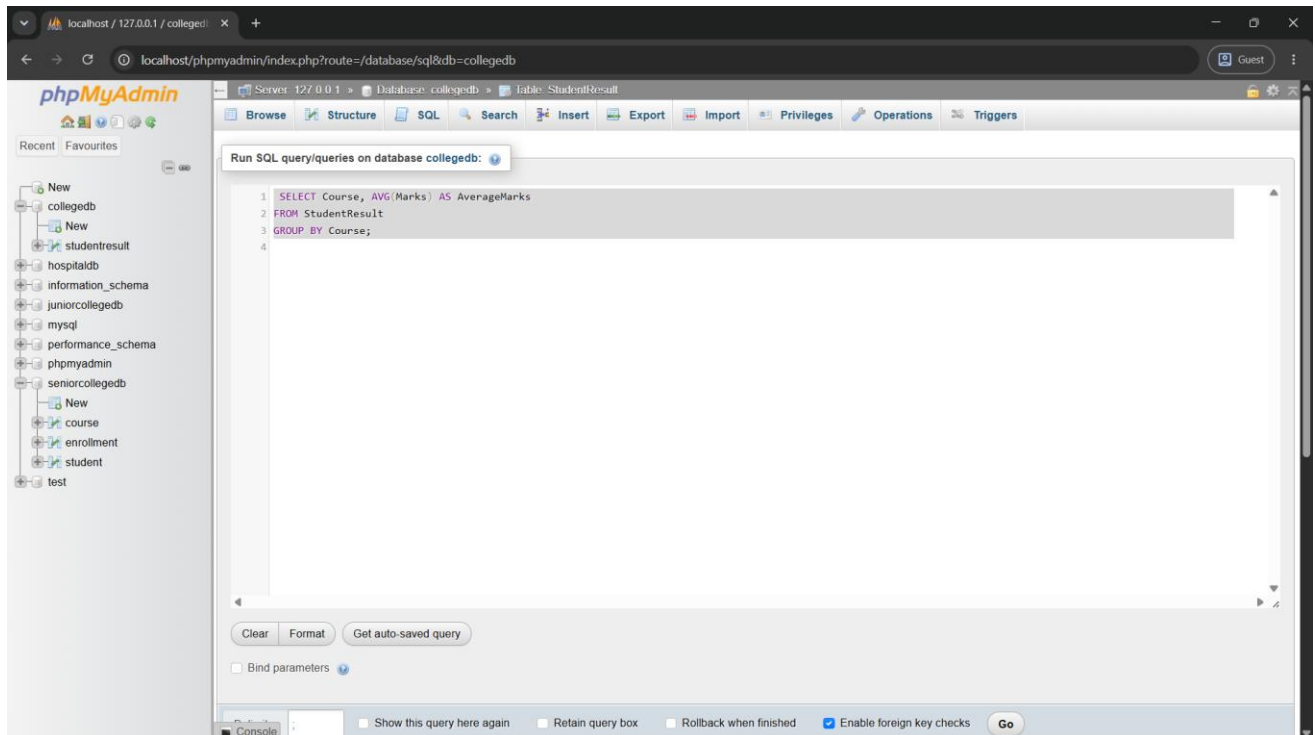
QUERY:

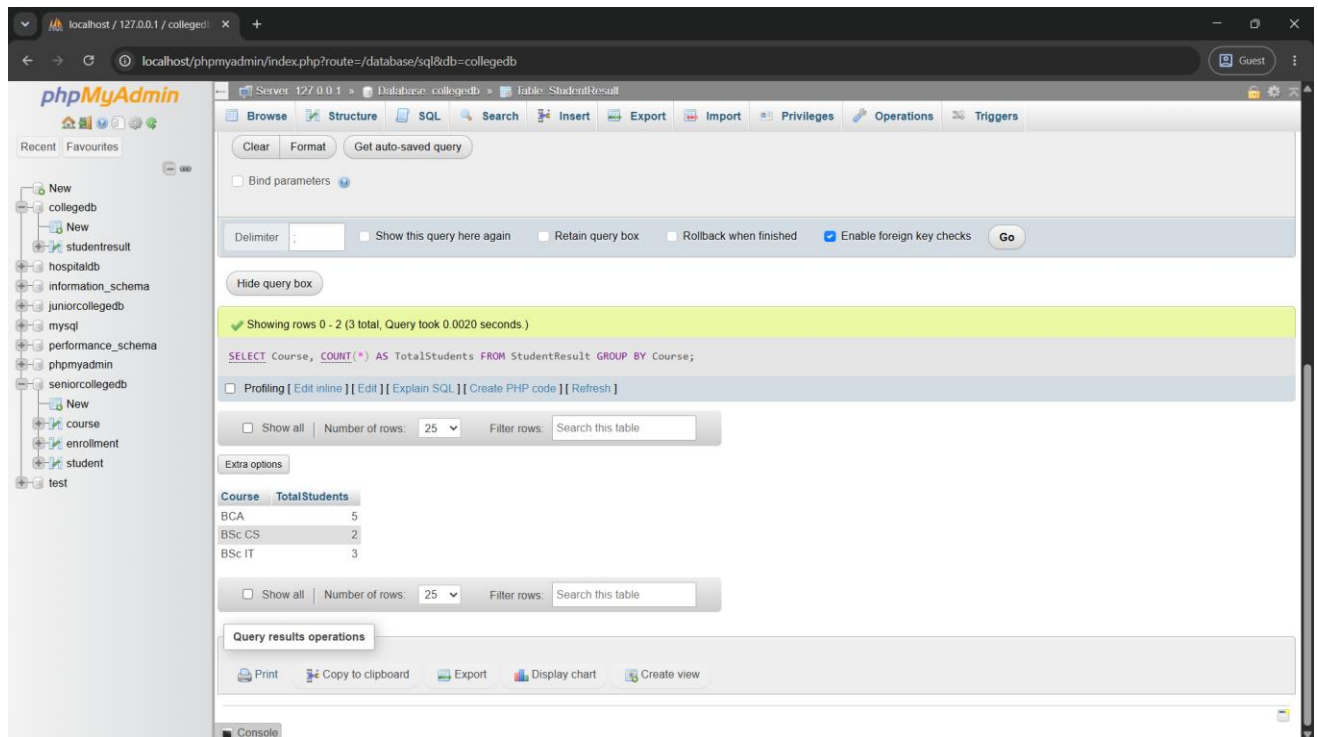
SELECT Course, AVG(Marks) AS AverageMarks

FROM StudentResult

GROUP BY Course;

OUTPUT:





Q5) Write a MySQL query to display only those courses where the number of students is greater than 2.

Use:

Clause
GROUP BY
HAVING

QUERY :

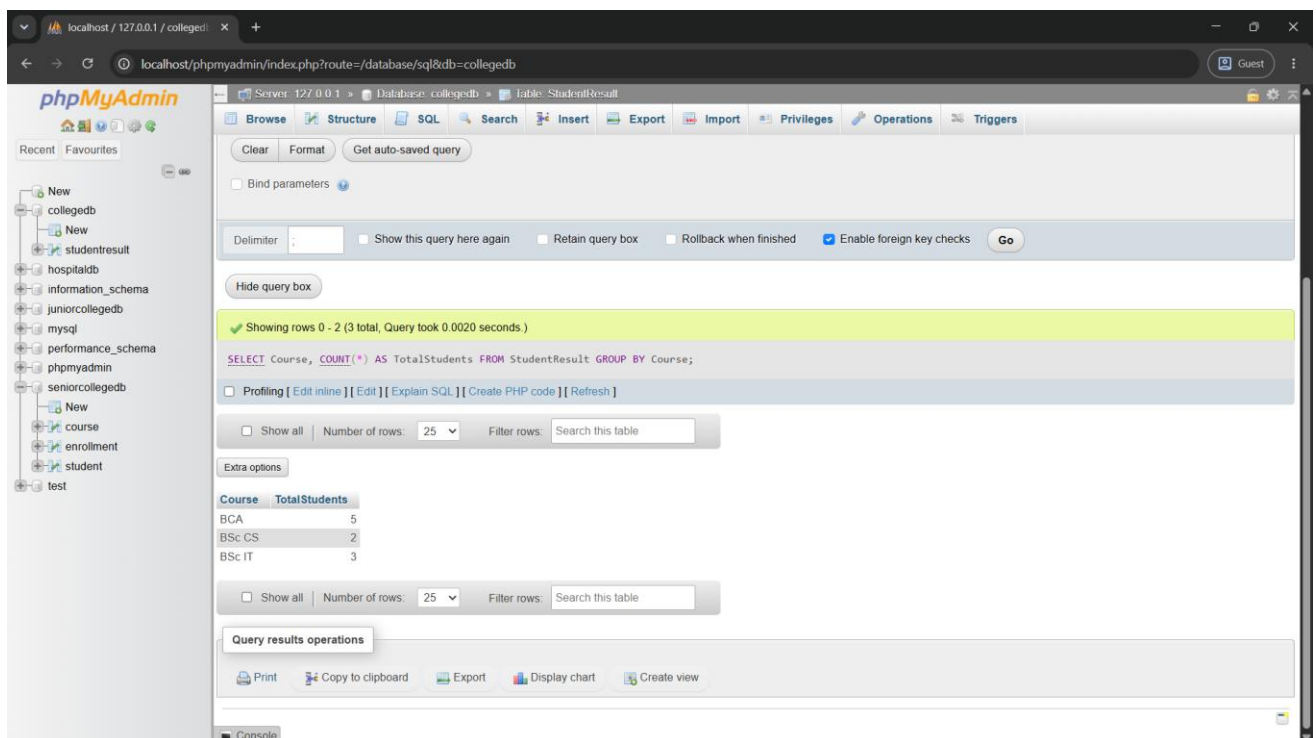
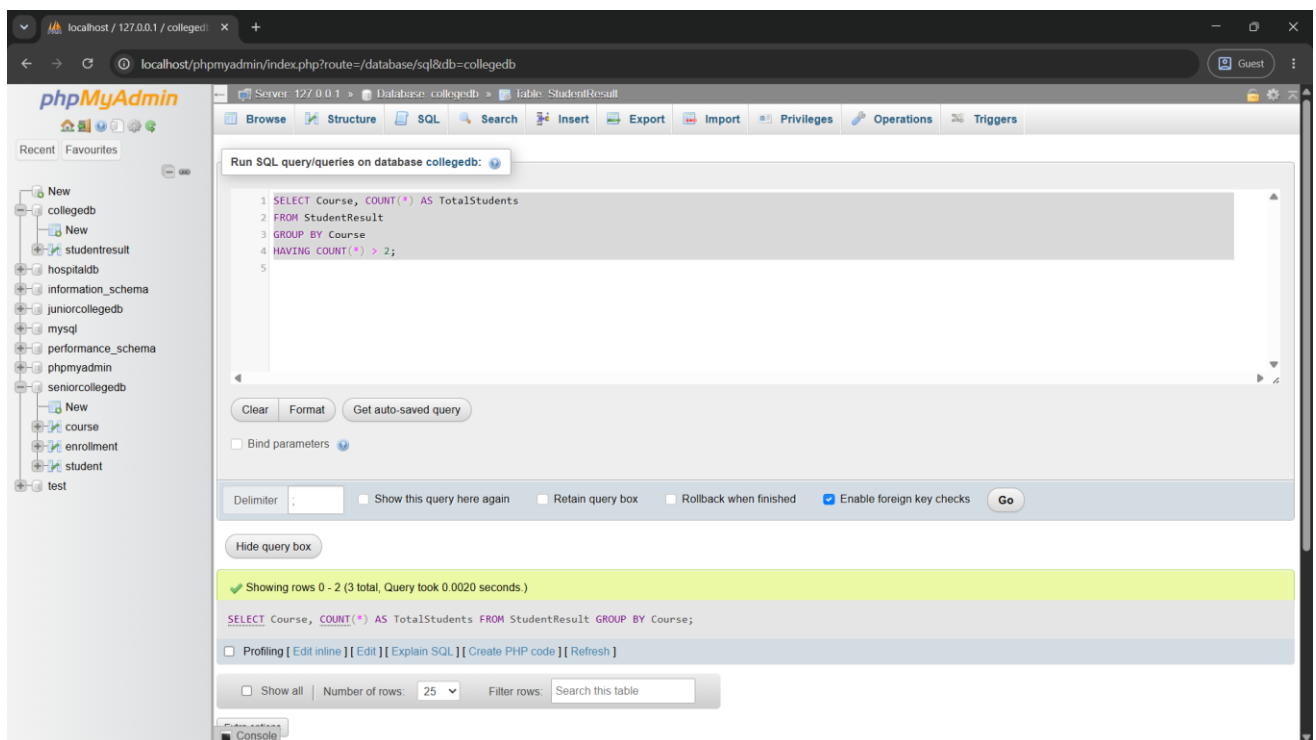
SELECT Course, COUNT(*) AS TotalStudents

FROM StudentResult

GROUP BY Course

HAVING COUNT(*) > 2;

OUTPUT:



Q6) Write a MySQL query to analyze course performance.

The output should display:

Column
Course

Column
TotalStudents
AverageMarks
LowestMarks
HighestMarks

Sort the result by `AverageMarks` in descending order.

QUERY:

SELECT

Course,

COUNT(*) AS TotalStudents,

AVG(Marks) AS AverageMarks,

MIN(Marks) AS LowestMarks,

MAX(Marks) AS HighestMarks

FROM StudentResult

GROUP BY Course

ORDER BY AverageMarks DESC;

OUTPUT:

localhost / 127.0.0.1 / collegedi

localhost/phpmyadmin/index.php?route=/database/sql&db=collegedb

phpMyAdmin

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- course
- enrollment
- student
- test

Run SQL query/queries on database collegedb:

```
2 Course,
3 COUNT(*) AS TotalStudents,
4 AVG(Marks) AS AverageMarks,
5 MIN(Marks) AS LowestMarks,
6 MAX(Marks) AS HighestMarks
7 FROM StudentResult
8 GROUP BY Course
9 ORDER BY AverageMarks DESC;
10
```

Clear Format Get auto-saved query

☐ Bind parameters

Delimiter ; ☐ Show this query here again ☐ Retain query box ☐ Rollback when finished ☒ Enable foreign key checks Go

Hide query box

Showing rows 0 - 2 (3 total, Query took 0.0019 seconds.)

```
SELECT Course, COUNT(*) AS TotalStudents, AVG(Marks) AS AverageMarks, MIN(Marks) AS LowestMarks, MAX(Marks) AS HighestMarks FROM StudentResult GROUP BY Course ORDER BY AverageMarks DESC;
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Console

localhost / 127.0.0.1 / collegedi

localhost/phpmyadmin/index.php?route=/database/sql&db=collegedb

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Bind parameters

Delimiter ; ☐ Show this query here again ☐ Retain query box ☐ Rollback when finished ☒ Enable foreign key checks Go

Hide query box

Showing rows 0 - 2 (3 total, Query took 0.0019 seconds.)

```
SELECT Course, COUNT(*) AS TotalStudents, AVG(Marks) AS AverageMarks, MIN(Marks) AS LowestMarks, MAX(Marks) AS HighestMarks FROM StudentResult GROUP BY Course ORDER BY AverageMarks DESC;
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

Course	TotalStudents	AverageMarks	LowestMarks	HighestMarks
BCA	5	84.8000	64	95
BSc IT	3	78.3333	69	90
BSc CS	2	77.0000	73	81

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Query results operations

Print Copy to clipboard Export Display chart Create view

Console